SS-C-218a
NOVEMBER 21, 1960
SUPERSEDING
Fed. Spec. SS–C–218
February 7, 1957

FEDERAL SPECIFICATION

CEMENT, SLAG

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. CLASSIFICATION

1.1 Types.—This specification covers the following two types of slag cement. When the type is not specified, type S shall be furnished.

Type S.—Slag cement for use as a blend with portland cement in making concrete, and as a blend with hydrated lime in making masonry mortar.

Type SA. — Air-entraining slag cement for the same uses as type S.

1.2 Definition.—For the purpose of this specification, slag cement is finely divided material consisting essentially of water-quenched granulated blast-furnace slag and hydrated lime. It shall consist of at least 60 percent water-quenched blast-furnace slag by weight. Water-quenched granulated blast-furnace slag is the nonmetallic product consisting essentially of silicates and aluminosilicates of calcium which is developed simultaneously with iron in a blast furnace and is produced by rapidly quenching the molten material with water. The hydrated lime shall meet the requirement for type M, of the specification for Hydrated Lime for Structural Purposes, SS–L–351, except that the unhydrated oxides shall not be over 8 percent.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

SS–L–351—Lime, Hydrated (for) Structural Purposes.

UU–S–48—Sacks, Shipping, Paper.

Federal Standards:

Fed. Test Method Std. No. 158 — Cements, Hydraulic; Sampling, Inspection and Testing.

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

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Military Standard:

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Chemical requirements.—Slag cement shall conform to the chemical test requirements prescribed in table I.

TABLE I.—*Chemical requirements, types S and SA*

Sulfur trioxide (SO_3) max., percent	4.0
Manganic Oxide (Mn_2O_3) max., percent	1.5
Sulfide sulfur (S) max., percent	2.0
Insoluble residue, max., percent	1.0
Loss on ignition, max., percent	4.0

3.2 Physical requirements. — Slag cement shall conform to the physical test requirements prescribed in table II.

TABLE II.—*Physical requirements*

	Type S	Type SA
Fineness, Air permeability		
Average value, min., sq. cm./g	4700	4700
Min. value, any one sample in average sq. cm./g	4200	4200
Soundness, Autoclave expansion or contraction, max., percent	0.50	0.50
Time of Setting, Gillmore apparatus		
Initial set, min., not less than	60	60
Final set, hours, not more than	12	12
Air content of mortar, percent	12 max.	19±3
Compressive strength, p.s.i.		
1 day in moist air, 6 days in water	600	500
1 day in moist air, 27 days in water	1500	1000

3.3 Slag composition. — The granulated blast-furnace slag shall conform to the following composition:

$$\frac{CaO + MgO + \frac{1}{3} Al_2O_3}{SiO_2 + \frac{2}{3} Al_2O_3} \geqq 1.0$$

Note.—Granulated blast-furnace slag having a chemical composition within the following ranges generally meets the requirements of the above formula:

Silicon dioxide (SiO_2), percent	30 to 40
Aluminum oxide (Al_2O_3), percent	8 to 18
Ferrous oxide (FeO), percent	0 to 1
Calcium oxide (CaO), percent	40 to 50
Magnesium oxide (MgO), percent	0 to 8
Manganic oxide (Mn_2O_3), percent	0 to 2
Sulfide sulfur (S), percent	0 to 2

3.4 Additions during grinding. — Water and/or untreated calcium sulfate may be used in amounts such that the limits shown in table I for ignition loss and sulfur trioxide shall not be exceeded. At the option of the manufacturer, nondeleterious materials other than water and/or calcium sulfate may be used as grinding aids in an amount less than 1 percent.

3.5 Air-entraining addition. — When air-entraining cement is specified, there shall be interground with the slag an addition of such a nature and in such an amount that the resulting product complies with the applicable requirements of this specification. At the request of the purchaser, the manufacturer shall furnish a written statement giving the source, amount, and brand name of the addition used. At the request of the purchaser, the manufacturer shall also furnish test data (note) acceptable to the purchaser showing that air-entraining portland cement that contained this addition and met the applicable requirements of tables I and II of Federal Specification SS–C–192 had a durability factor of not less than 80 when tested in accordance with the procedures of method 4201.

Note.—Test data showing that an air-entraining addition complies with the requirements of ASTM Designation C–226 will be regarded as acceptable.

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3.6 Manufacturers statement.—At the request of the purchaser, the manufacturer shall state in writing the source, amount, and composition of the granulated blast-furnace slag used in the finished cement and the proportions of slag and lime in the slag cement. The composition of the finished cement shall conform to that shown in the statement within the following tolerances: SiO_2 ± 3 percent, Al_2O_3 ± 2 percent, CaO ± 3 percent.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The applicable methods outlined in Federal Test Method Standard No. 158 shall be followed to determine compliance with the provisions of this specification.

4.2 Chemical analysis.—Chemical analysis shall be made in accordance with applicable methods outlined in Federal Test Method Standard No. 158. Referee methods shall be used in case of dispute, or when the results so obtained by optional methods indicate a result so close to the specification limit that meeting or failing to meet the specification requirement is in doubt. When referee methods are required all determinations of constituents involved in the computations shall be made entirely by referee methods and the separations prior to these methods shall be made by referee methods.

4.3 Fineness. — Slag cement has a lower apparent specific gravity than portland cement. The weight of the sample shall be adjusted so that a firm, hard bed is formed by the compacting process. Calculate the surface using the apparent specific gravity of the slag cement being tested. Determine the specific gravity by means of the standard Le Chatelier flask with water-free kerosene as the liquid medium.

4.4 Autoclave soundness test.—The autoclave soundness test shall be made by the procedure outlined in Federal Test Method Standard No. 158, except that the specimens shall remain in the moist cabinet for 48 ± ½ hours before autoclaving.

4.5 Air content of mortar.—Calculate the air content of the mortar using the specific gravity of the slag cement being tested.

4.6 Cause for rejection.—The cement shall be rejected if it fails to meet any of the requirements of this specification. Cement remaining in storage for a period of more than six months after completion of test may be retested and shall be rejected if it fails to conform to any of the requirements of this specification. Cement having an air content between 12 and 16 percent as determined by the air-entrainment test may be accepted at the discretion of the purchaser when air-entraining concrete is to be used and the air content of the concrete is controlled at the mixer.

5. PREPARATION FOR DELIVERY

5.1 For civil agency procurement, the levels of preservation, packaging and packing used are defined in Federal Standard No. 102.

5.2 Packaging.—The cement shall be delivered in bags, barrels, or in bulk, as specified. A bag shall contain 0.48 cubic foot of material by absolute volume. All packages shall be in good condition at the time of inspection. Packages varying more than 5 percent from the specified weight may be rejected; and if the average weight of packages in any shipment, as shown by weighing 50 packages at random, is less than that specified, the entire shipment may be rejected. When shipping in bags, packing shall be in accordance with one of the following levels, as specified.

5.2.1 *Level A.*—Cement in quantities of 94 pounds net shall be packed in bags conforming to Federal Specification UU–S–48, X series.

5.2.2 *Level B.*—Cement in quantities of 94 pounds net shall be packed in bags conforming to Federal Specification UU–S–48, L/W series.

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5.2.3 *Level C.*—Cement shall be packed in a manner which will assure arrival at destination in satisfactory condition and which will be acceptable to the carrier at the lowest transportation cost.

5.3 Marking.

5.3.1 *Civil agencies.*—Marking shall be in accordance with Federal Standard No. 123. When cement is delivered in packages, the name and brand of the manufacturer, the words "Slag Cement", the type of slag cement, the net weight and absolute volume shall be plainly indicated thereon. Similar information shall be provided in the shipping advices accompanying the shipment of packaged or bulk cement.

5.3.2 *Military procurement.* — Marking shall be in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Ordering data. — Purchasers should specify the type of cement required and should exercise any desired options offered herein (1.1, 5.1, 5.2).

6.2 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.3 Federal Specifications do not include all types, classes, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, classes, etc., which are suitable for Federal Government requirements.

6.4 Transportation description.—Transportation description applicable to this item is:

Slag, not otherwise indexed by name, granulated or lump.
Carload minimum weight 60,000 pounds.
Truckload minimum weight 40,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:
Navy—Y.

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